

lec 37.

Convergence in mean square sense

Sequence of RVs . $X_1, X_2, \dots$

$X_n \longrightarrow X$ in mean square sense

$$\lim_{n \rightarrow \infty} E[|X_n - X|^2] = 0.$$

① Convergence in mean square sense
 $\Rightarrow$ convergence in probability

$$P(|X_n - X| > \varepsilon) = P((X_n - X)^2 > \varepsilon^2).$$

Markov
inequality
for $Y = (X_n - X)^2$

$$\frac{E[Y]}{\varepsilon^2} = \frac{E[(X_n - X)^2]}{\varepsilon^2}$$

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) \leq \lim_{n \rightarrow \infty} \frac{E[(X_n - X)^2]}{\varepsilon^2} = 0.$$

② Convergence in a.s. sense $\not\Rightarrow$ convergence in mean square sense.

$$X_n = \begin{cases} 0 & \text{w.p. } 1 - \frac{1}{n^2} \\ n^2 & \text{w.p. } \frac{1}{n^2} \end{cases}$$

$$X = 0 \quad \text{w.p. } 1.$$

$$\begin{aligned} E[(X_n - X)^2] &= E[X_n^2] \\ &= 0^2 \times \left(1 - \frac{1}{n^2}\right) + (n^2)^2 \times \frac{1}{n^2} \\ &= n^2. \end{aligned}$$

$$\lim_{n \rightarrow \infty} E[(X_n - X)^2] \neq 0.$$

$$\text{let } A_n = \{\omega : |X_n(\omega) - X(\omega)| > \varepsilon\}.$$

$$P(A_n) = P(|X_n| > \varepsilon) = \frac{1}{n^2}.$$

$$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

$$\Rightarrow P(\limsup A_n) = 0$$

from Borel Cantelli lemma.

$$\Rightarrow X_n \xrightarrow{a.s.} X$$

③ Borel - Cantelli lemma 2

Let $A_1, A_2, \dots$ be independent events st

$$\sum_n P(A_n) = \infty. \text{ then}$$

$$P(\limsup A_n) = 1$$

For independence to hold

$$P(A_1, \dots, A_k) = \prod_{i=1}^k P(A_i)$$

for any k

Proof: $P(\limsup A_n) = P\left(\bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n\right)$

$$B_1 = \bigcup_{n=1}^{\infty} A_n$$

$$= P\left(\bigcap_{N=1}^{\infty} B_N\right)$$

$$B_2 = \bigcup_{n=2}^{\infty} A_n$$

:

$$B_1 \supseteq B_2 \supseteq \dots$$

Corollary 2 of continuity property

$$= \lim_{N \rightarrow \infty} P(B_N)$$

$$= \lim_{N \rightarrow \infty} P\left(\bigcup_{n=N}^{\infty} A_n\right)$$

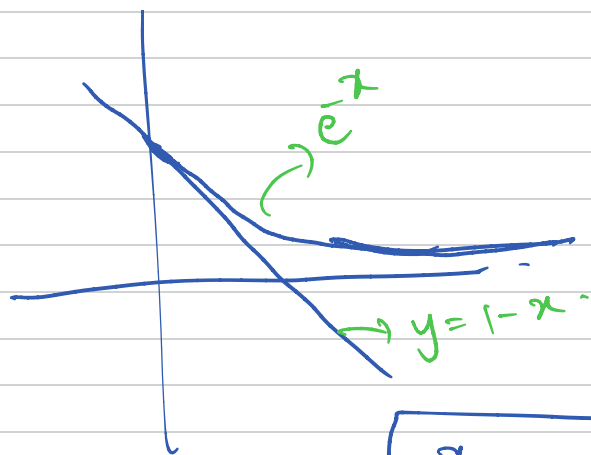
Continuity property

$$= \lim_{N \rightarrow \infty} \lim_{M \rightarrow \infty} P\left(\bigcup_{n=N}^M A_n\right)$$

$$P\left(\bigcup_{n=N}^M A_n\right) = 1 - P\left(\bigcap_{n=N}^M A_n^c\right).$$

$$= 1 - \prod_{n=N}^M P(A_n^c).$$

$$= 1 - \prod_{n=N}^M (1 - P(A_n))$$



$$y = 1-x. \quad \boxed{e^{-x} \geq 1-x}$$

$$-e^{-x} \leq -(1-x).$$

We know that $\sum_{n=1}^{\infty} P(A_n) = \infty$.

$$\hat{S}_n = \sum_{i=n}^{\infty} P(A_i).$$

$$\geq 1 - \prod_{n=N}^M e^{-P(A_n)}.$$

$$= 1 - e^{-\sum_{n=N}^M P(A_n)}$$

$$P\left(\bigcup_{n=N}^M A_n\right) \geq 1 - e^{-\sum_{n=N}^M P(A_n)}.$$

$$\lim_{M \rightarrow \infty} P\left(\bigcup_{n=N}^M A_n\right) \geq 1 - e^{-\sum_{n=N}^{\infty} P(A_n)}.$$

$$= 1 - e^{-\infty} = 1.$$

$$P(\limsup A_n) = \lim_{N \rightarrow \infty} \lim_{M \rightarrow \infty} P\left(\bigcup_{n=N}^M A_n\right) \geq 1.$$

$$\Rightarrow P(\limsup A_n) = 1.$$

④ Convergence in mean square sense.

~~$\Rightarrow$~~ Convergence in a.s. sense.

$$X_n = \begin{cases} 0 & \text{w.p. } 1 - \frac{1}{n} \\ 1 & \text{w.p. } \frac{1}{n} \end{cases}$$

X_n 's are independent of each other.

Let $x = 0$ w.p. 1.

$$\begin{aligned} E[(X_n - x)^2] &= E[X_n^2] \\ &= (1)^2 \times \frac{1}{n} + 0^2 \times \left(1 - \frac{1}{n}\right) \\ &= \frac{1}{n}. \end{aligned}$$

$$\lim_{n \rightarrow \infty} E[(X_n - x)^2] = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

$\Rightarrow X_n \rightarrow x$ in mean square sense

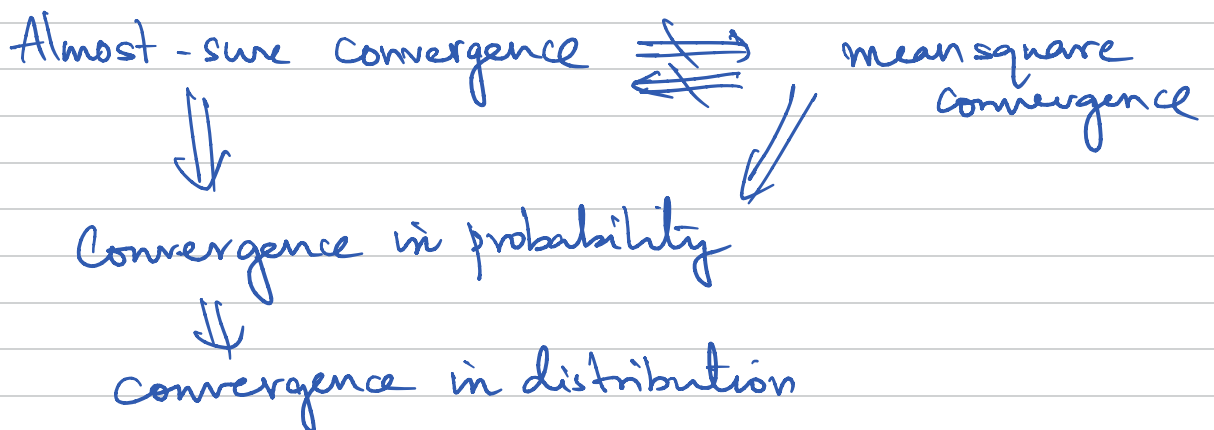
$$A_n = \{ |X_n - x| > \varepsilon \} = \{ |X_n| > \varepsilon \}$$

A_n are independent
as X_n 's are independent

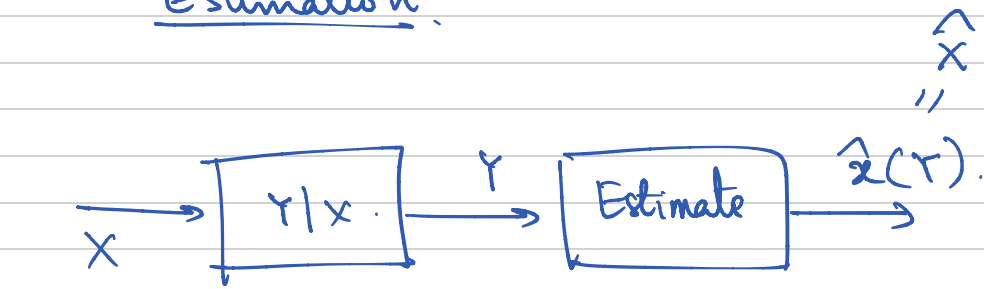
$$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

$$\Rightarrow P(\limsup A_n) = 1$$

Borel Cantelli lemma 2.



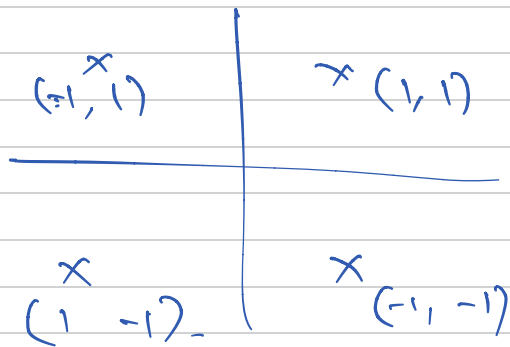
Estimation.



prior distribution
①

Probability $P(\hat{x} \neq x)$
of error in estimation.

This metric is relevant when x is a discrete R.V.



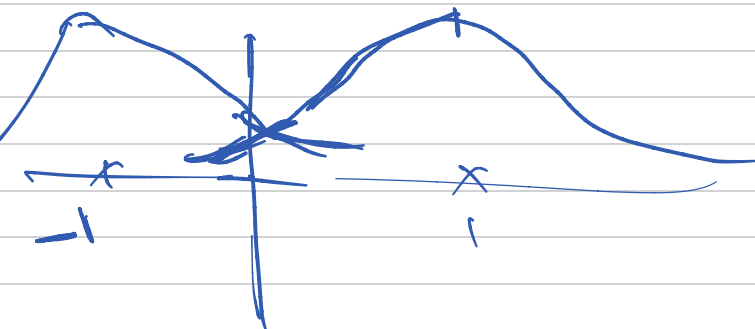
$$(x_1, x_2) \in \{(-1, 1), \dots\}$$

$$(z_1, z_2) \sim N(0, \sigma^2, 0, \sigma^2, 0)$$

$$(Y_1, Y_2) = (x_1 + z_1, x_2 + z_2)$$

QAM.

Digital Communication



$$X \in \{+1, -1\}$$

$$Z \sim N(0, \sigma^2)$$

From $Y = X + Z$
estimate x .

② Continuous R.V. x .

Mean Square Error.
for continuous R.V.
 $E[(\hat{x} - x)^2] = \text{MSE}$.

① Estimating a discrete R.V. from Y .

MAP estimator
(Maximum A posteriori Probability)

$$\hat{x}_{\text{MAP}}(y) = \arg \max_{x \in \mathcal{X}_0} P_{x|Y}(x|y).$$

y discrete $\swarrow$

$$= \arg \max_{x \in \mathcal{X}_0} \frac{P_{Y|X}(y|x) P_X(x)}{P_Y(y)}$$

$$f_{Y|X}(y|x) P_X(x).$$

$$\leq f_{Y|X}(y|\hat{x}_{\text{MAP}}(y)) P_X(\hat{x}_{\text{MAP}}(y)). \quad \text{or} \quad \arg \max_{x \in \mathcal{X}_0} \boxed{\frac{f_{Y|X}(y|x) P_X(x)}{f_Y(y)}}.$$

MAP minimizes the probability of estimation error.

Proof: $P(X \neq \hat{X}) \geq P(X \neq \hat{X}_{\text{MAP}}).$

$$\hat{X} = \hat{x}(Y)$$

$$\hat{X}_{\text{MAP}} = \hat{x}_{\text{MAP}}(Y)$$

$$\iff P(X = \hat{X}) \leq P(X = \hat{X}_{\text{MAP}}).$$

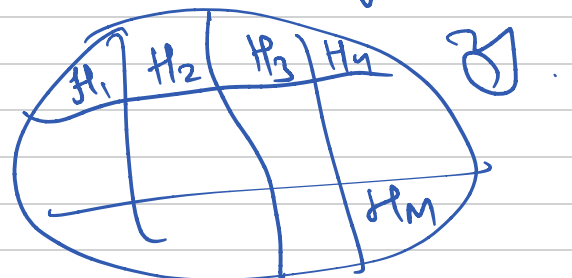
$$P(X = \hat{X}) = \sum_{x \in \mathcal{X}_0} P(X=x) P(X = \hat{X} | X=x)$$

Any estimator $\hat{x}(Y)$

does the following. It divides the space of Y .

into $M = |\mathcal{X}_0|$ parts.

If $y \in \mathcal{H}_x$
then $\hat{x}(y) = x$.



$$P(X = \hat{x}) = \sum_{x \in \mathcal{X}_0} P_X(x) P(\hat{x}(Y) = x | X=x)$$

assuming Y is continuous $\leftarrow$ $= P(Y \in H_x)$.

$$= \sum_{x \in \mathcal{X}_0} P_X(x) \int_{y \in H_x} f_{Y|X}(y|x) dy$$

$$y \in H_x \iff \hat{x}(y) = x$$

$$= \sum_{x \in \mathcal{X}_0} \int_{y \in H_x} P_X(\underline{x}) f_{Y|X}(y|\underline{x}) dy$$

$$= \sum_{x \in \mathcal{X}_0} \int_{y \in H_x} P_X(\hat{x}(y)) f_{Y|X}(y|\hat{x}(y)) dy$$

$$= \int_{y=-\infty}^{\infty} P_X(\hat{x}(y)) f_{Y|X}(y|\hat{x}(y)) dy$$

$$\leq \int_{y=-\infty}^{\infty} P_X(\hat{x}_{\text{MAP}}(y)) f_{Y|X}(y|\hat{x}_{\text{MAP}}(y)) dy$$

$$= P(X = \hat{x}_{\text{MAP}}(Y)).$$